

PAPER_475 — UQFF Sub-Term Physics Modules Catalogue (44 Standalone Calculators)

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Abstract

This whitepaper catalogues 44 standalone sub-term physics calculator modules that constitute the atomic building blocks of the UQFF and MUGE frameworks. Each module encapsulates a single physical quantity — from aether vacuum densities and buoyancy coupling constants to solar cycle frequencies and stellar surface temperatures — as a self-contained C++ class. This catalogue provides a reference index for module equations, key parameter values, output ranges, and integration targets within MAIN_1_CoAnQi.cpp and CondensedPhysics.py.

1. Introduction

The 44 sub-term modules were extracted from grok_share_b0a3dc1d.txt (10,420 lines) and represent the granular physics underpinning the UQFF unified field equation:

$$F_{U,Bi,i} = \int [\text{Ug coupling} \cdot \text{vacuum terms} \cdot \text{buoyancy} \cdot \text{aether} \cdot \text{stellar params}] dx$$

Each module is implemented as a C++ class in modules/subterms/ with standard interface: computeX(), updateVariable(), getEquationText(), printVariables().

2. Module Catalogue

Category A: Vacuum Energy Modules

#	Module	Key Equation	Key Value
1	AetherVacuumDensityModule	$\rho_A = E_A/c^2$	$7.09 \times 10^{-36} \text{ J/m}^3$
2	UniversalInertiaVacuumModule	[energy]	$7.09 \times 10^{-36} \text{ J/m}^3$
3	ScmVacuumDensityModule	$\rho_{vac_SCm} = \rho_{UA}/10$	$7.09 \times 10^{-37} \text{ J/m}^3$
4	UaVacuumDensityModule	$\rho_{UA} = E_{UA}/c^2$ [mass]	$7.88 \times 10^{-53} \text{ kg/m}^3$
5	BackgroundAetherModule	$\rho_A = (\rho_A/c^2)\partial_\mu\phi$	$\rho_A = 7.09 \times 10^{-36} \text{ J/m}^3$

Category B: Coupling Constants

#	Module	Key Equation	Key Value
6	AetherCouplingModule	$\eta_{\mu\nu} = g_{\mu\nu} + \eta T_s^{\mu\nu}$	$\eta \approx 1/E_{s_total} \approx 1e-15$
7	UgCouplingModule	k_i weights Ug_i	$k_1=1.5, k_2=1.2, k_3=1.8, k_4=1.0$

#	Module	Key Equation	Key Value
8	BuoyancyCouplingModule	$U_{bi} = -\beta_i U_{gi} \Omega_g \dots$	$\beta_i = 0.6$ uniform
9	InertiaCouplingModule	$I = M r^2 \Omega^2/c^2$	Sun: $\approx 7.8 \times 10^{-10}$
10	UgIndexModule	$Ug_{\{i,n\}} = G M/r^2 Q_{i,n}$	4×26 array

Category C: Solar/Stellar Parameters

#	Module	Key Equation	Key Value
11	SolarWindBuoyancyModule	$\epsilon_{sw} = 0.001 \times (1 + A \sin(\omega t))$	$\epsilon_{sw} = 0.001$ baseline
12	SolarWindModulationModule	$v_0(t) = v_0(1 + A \sin(\omega t))$	$v_0 = 400$ km/s, $A = 0.2$
13	SolarWindVelocityModule	$v \in [400, 800]$ km/s	$v_{fast} = 750$ km/s
14	SolarCycleFrequencyModule	$f_{sc} = 1/(11 \text{ yr})$	2.88×10^{-9} Hz
15	HeliosphereThicknessModule	$r_{HP} - r_{TS}$	~ 30 AU ($\sim 4.5 \times 10^{12}$ m)
16	StellarMassModule	$M_s(t) = M_0(1 - \gamma_{ML} t)$	$\gamma_{ML} \approx 1.4 \times 10^{-14} \text{ s}^{-1}$
17	StellarRotationModule	$\omega_s = 2\pi/P_{rot}$	Sun: 2.87×10^{-6} rad/s
18	SurfaceMagneticFieldModule	$B_{pole} = M_{mag}/4\pi r^3$	Magnetar: 4.4×10^{13} T
19	SurfaceTemperatureModule	$T_e = (L/4\pi\sigma r^2)^{0.25}$	Sun: 5778 K
20	MagneticMomentModule	$\mu = I A_{vort}$	Solar: $\sim 1 \times 10^{21} \text{ A} \cdot \text{m}^2$

Category D: Galactic/Astrophysical Parameters

#	Module	Key Equation	Key Value
21	GalacticDistanceModule	$r_g = \text{virial radius}$	MW: 2.55×10^{20} m
22	GalacticSpinModule	$\Omega_g = 2\pi/T_{gal}$	MW: 7.3×10^{-16} rad/s
23	GalacticBlackHoleModule	$M_{BH} \propto \sigma^4 (M - \sigma)$	Sag A*: 8×10^{36} kg
24	FeedbackFactorModule	$f_{env} = f_{AGN} + f_{SN} + f_{SF}$	$f_{AGN} = 0.1$, $f_{SN} = 0.05$
25	MagneticStringModule	$T_s = (\mu_0 I^2/4\pi) \ln(L/a)$	$\sim 10^{28}$ N (cosmic string)
26	Ug3DiskVectorModule	$Ug3_{disk} = G M_{disk}/r^2(h/r)$	MW: $h/r \approx 0.07$
27	Ug1DefectModule	$Ug1_{corr} = Ug1(1 - \delta_{def})$	$\delta_{def} \approx 0.05 - 0.15$

Category E: Quantum & Field Calculators

#	Module	Key Equation	Key Value
28	DPMModule	26-sphere: $(x-h)^2 + \dots = r^2$	$SCm E = 10^{42}$ J
29	UnifiedFieldModule	$F_U = \sum k_i U_{g_i} + F_{bi} \dots$	Orchestrator
30	StressEnergyTensorModule	$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + pg_{\mu\nu}$	Trace = $-\rho + 3p$
31	QuasiLongitudinalModule	$\epsilon_{QL} = \epsilon_0 E^2/2$	$\sim 1e-12 \text{ J/m}^3$
32	OuterFieldBubbleModule	$r_{ro} \exp(H t)$	At 10 Gyr: $r = 1.28 r_0$
33	ReciprocatationDecayModule	$\rho_{rec} \exp(-t/\tau_{rec})$	$\tau_{rec} \sim 1$ Gyr
34	ScmPenetrationModule	$\lambda_{SCm} = \lambda(\rho/\rho_{SCm})^{0.5}$	London-analog depth
35	ScmReactivityDecayModule	$d[SCm]/dt = -k_r [SCm]$	[SSq]=0.57, $k_r \sim 1e-18 \text{ s}^{-1}$
36	ScmVelocityModule	$v_{SCm} = c/n_{SCm}$	$n_{SCm} \approx 1.0000001$
37	PiConstantModule	$\pi = 4\sum (-1)^k/(2k+1)$	Leibniz series
38	CorePenetrationModule	$\delta = (\rho_{core}/\rho_{avg})^n r_{core}$	NS core: $\delta \rightarrow 0$
39	NegativeTimeModule	$g(t < 0) = g_0 \cos(\omega t)$	$f_{TRZ} = 0.1$
40	TimeReversalZoneModule	$TRZ = r_{TRZ}/r$	Canonical: 0.1
41	HeavisideFractionModule	$f = 0 (f < 0), 1 (f \geq 0)$	Threshold: 0
42	StepFunctionModule	$\theta(x)$: boxcar windows	Regime switching

Category F: Index / Utility Modules

#	Module	Key Equation	Key Value
43	GalacticBlackHoleModule	$\gamma_{BH} = G M_{BH}/r^2$	At $r=5.5 \times 10^{10}$ m (Sag A*)
44	InertiaCouplingModule	$\gamma_d = 1 + I_c$	Ug3 relativistic boost

3. Integration Targets

3.1 MAIN_1_CoAnQi.cpp

Sub-term modules are candidates for batch extraction as PhysicsTerm subclasses:

```
// Example: ScmVacuumDensityModule → ScmVacuumDensityTerm
class ScmVacuumDensityTerm : public PhysicsTerm {
public:
    double compute(const SystemParams& params) override {
        return 7.09e-37; // ρ_vac_SCm [J/m³]
    }
    std::string getName() const override { return "SCm Vacuum Density"; }
};
```

Target integration point: **Batch 24** in SOURCE1-116 registry near line 6688+.

3.2 CondensedPhysics.py

Each sub-term module maps to a Calculator class method:

```
class SubTermCalculator:
    def compute_aether_vacuum_density(self) -> dict:
        # Returns ρ_vac_A = 7.09e-36 J/m³ with equation text
```

Target: new UQFFSubTermCalculator in CondensedPhysics.py Part 5 (after line 60000).

3.3 index.js

Module constants exportable as:

```
const SUB_TERMS = {
    rho_vac_UA: 7.09e-36, // J/m³
    rho_vac_SCm: 7.09e-37, // J/m³
    beta_i: 0.6, // buoyancy coupling
    kappa: 0.0005/86400, // per-second DPM rate
    SSq: 0.57, // SCm calibration
};
```

4. Calibration Constants Summary

Constant	Value	Source Module
κ	0.0005/day	DPMModule
[SSq]	0.57	ScmReactivityDecayModule
β _i	0.6 (all i)	BuoyancyCouplingModule

Constant	Value	Source Module
ε_{sw}	0.001	SolarWindBuoyancyModule
η	$\sim 10^{-15}$	AetherCouplingModule
H_{SCm}	0.99	ScmVacuumDensityModule
k_1, k_2, k_3, k_4	1.5/1.2/1.8/1.0	UgCouplingModule
f_{TRZ}	0.1	TimeReversalZoneModule

5. Scientific Context

The 44 sub-term modules collectively represent a field-theoretic decomposition of gravity that accounts for:

- **Vacuum energy hierarchy:** $\rho_{\text{UA}} : \rho_{\text{SCm}} : \rho_{\text{cosm}} = 10 : 1 : 0.001$
- **Coupling hierarchy:** k_3 (string/rotation) $> k_1$ (dipole) $> k_2$ (charge) $> k_4$ (vacuum)
- **Time evolution:** DPM birth model $\rightarrow$ inflation $\rightarrow$ SCm decay $\rightarrow$ solar wind modulation $\rightarrow$ present
- **Multi-scale validity:** Sub-parsec (heliosphere) to Gpc (galaxy clusters)

6. Conclusion

This catalogue documents 44 atomic physics calculator modules that implement the sub-term physics of the UQFF/MUGE framework. The modules span vacuum energy, coupling constants, stellar parameters, galactic scales, and quantum field calculators. They constitute the modules/subterms/ library and are ready for integration as PhysicsTerm subclasses in MAIN_1_CoAnQi.cpp or as Python calculator methods in CondensedPhysics.py.

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_{T} (QED synchrotron)	UQFF U_{m} scattering kernel: $\sigma_{\text{T}} = 6.6524 \times 10^{-29} \text{ m}^2$	$\sigma_{\text{T}} = 6.6524 \times 10^{-29} \text{ m}^2$ (PDG QED exact)	PDG 2024	100% (exact QED input)
Astrophysical system luminosity X-ray / Radio	UQFF MUGE $g_{\text{total}} \rightarrow L_{\text{X}}$ via Stefan-Boltzmann + buoyancy flux: $L_{\text{X}} \approx g_{\text{total}} \times M_{\text{env}}$	$L_{\text{X}} L \geq 10^{37} \text{ erg/s}$	Chandra CXC	□ Consistent order of magnitude
GR Schwarzschild limit	UQFF g_{total} must satisfy $g \leq c^2/(2r_{\text{s}})$ at event horizon	$r_{\text{s}} = 2GM/c^2$ (GR exact)	PDG 2024 / GR	□ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$	Observed X-ray variability τ_{obs} (instrument monitoring)	Chandra CXC	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Astrophysical system through vacuum buoyancy coupling — a mechanism absent

from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

Source: grok_share_b0a3dc1d.txt L1502-9356 (44 class definitions)

Header index: modules/subterms/sub_terms_index.h

Tags: sub-terms, catalogue, vacuum-density, coupling-constants, solar-parameters, UQFF, MUGE, integration